

# Catalogue of AI Tests and Controls

Version 0.4 – June 2026

The Catalogue indexes testing solutions and control checklists designed to evaluate the trustworthiness and regulatory compliance of AI systems. It is open to contributions and encompasses:

- A **public website** where anyone can browse, explore, and submit solutions.
- A **downloadable version** from the Sandbox Configurator's GitHub repository that runs locally in a browser, enabling one-click plugin installation directly into the host environment. This version is not available yet and will be released soon.

Each solution in the Catalogue is integrated into the Sandbox Configurator through a **plugin**, i.e. a self-contained adapter that encapsulates the solution's execution logic and maps its outputs to the system's shared data model. Solutions listed in the Catalogue that have not yet been implemented as plugins serve as open contribution requests, inviting the community to develop them.

The catalogue will also include datasets in future releases; however, the current release focuses exclusively on plugins.

## 1. Catalogue structure: Tests and Controls

The catalogue holds two distinct entry types, and each plugin implements exactly one of them (declared in the Verification type field of Section 3):

- **Test:** executes a technical evaluation of an AI system and returns metrics (data type, range, unit; see Section 4.3).
- **Control:** a compliance checklist whose questions are assessed against regulatory requirements; it returns a readiness judgement rather than metrics, and each question is mapped to the EU AI Act provision(s) it addresses.

The two types share only their core descriptive fields (Name, Description, Project link). Their classification fields differ, and a control is not a test with an added Source. In particular, the Provider field is interpreted differently for each: for a test it is the developer or vendor (e.g. ETH Zurich, IBM) and is purely descriptive, whereas for a control it is the issuing authority (e.g. AESIA) and is surfaced as the Source classification tag (A6). The table below summarises which fields apply to each type:

| Field                | Tests                 | Controls               |
|----------------------|-----------------------|------------------------|
| Verification targets | Testing taxonomy (A1) | Controls taxonomy (A2) |
| Target systems (A3)  | ✓                     | —                      |
| Sector (A4)          | ✓                     | —                      |

|              |                                  |                                            |
|--------------|----------------------------------|--------------------------------------------|
| License (A5) | ✓                                | —                                          |
| Provider     | Developer / vendor (descriptive) | Issuing authority → becomes the Source tag |

## 2. Adding a Plugin

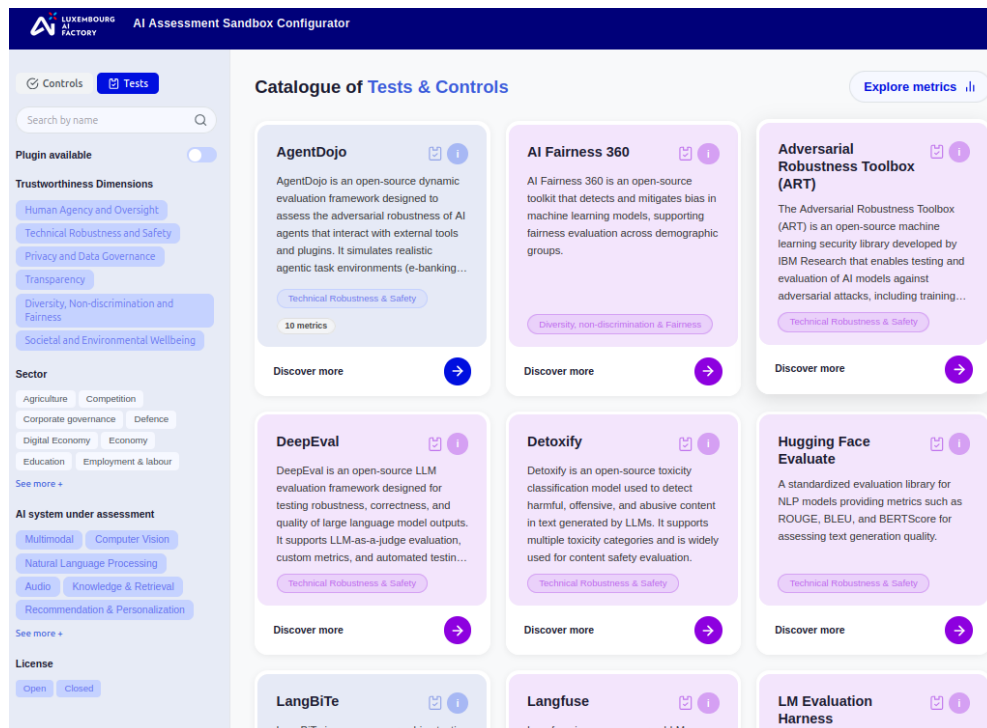
To contribute a plugin implementation, contributors must:

- Write metadata describing the solution, including tags to aid discoverability.
- Map the solution's outputs to the project's shared data model, i.e. the harmonisation layer used across the Sandbox Configurator to represent test and control check results in a consistent, DBMS-agnostic format.
- Implement the required plugin hooks in accordance with the Plug-in documentation in this Github repository: <https://github.com/lux-ai-factory/rfc>.

All submissions undergo a vetting process overseen by project administrators before becoming available for installation. Providers may additionally declare which regulatory requirements their solution addresses. As providers may naturally tend towards broader claims, one avenue worth exploring would be to involve a National Competent Authority or the AI Office in validating declared coverage, though the feasibility of such an approach remains to be assessed.

## 3. Discovering a Plugin

Users can filter plugins by category using tag, as shown in Figure 1.



**Figure 1. User interface of the catalogue. Users can explore plugins using the tags on the right that serve as filters.**

## 4. Plugin metadata

The following metadata template is used to describe and register a new plugin in the catalogue. For fields based on tags, the list of all available tags is reported in the appendix.

| Field                                  | Format                 | Explanation                                                                                       |
|----------------------------------------|------------------------|---------------------------------------------------------------------------------------------------|
| <b>Name</b>                            | <i>Text</i>            | Commercial name of plugin.                                                                        |
| <b>Description</b>                     | <i>Text</i>            | Brief overview of the testing solution, including its purpose, scope, and main functionalities.   |
| <b>License</b>                         | [TAG]                  | Type of license governing the use and distribution of the solution.                               |
| <b>Verification type</b>               | Technical test/control | Specifies the type of verification activity performed.                                            |
| <b>Provider</b>                        | <i>Text</i>            | Name of the organisation, team, or entity responsible for developing or maintaining the solution. |
| <b>Project</b>                         | Hyperlink              | Link to the associated project page, repository, or official documentation.                       |
| <b>Branch</b>                          | <i>Text</i>            | Development branch or implementation stream.                                                      |
| <b>Version</b>                         | <i>Text</i>            | Version number of the testing solution.                                                           |
| <b>Project maturity</b>                | <i>Text</i>            | Current maturity level (e.g., concept, prototype, validated in lab, deployed, maintained).        |
| <b>[OPTIONAL] Scientific reference</b> | <i>APA Text</i>        | Reference to related scientific publication(s) in APA format, if applicable.                      |
| <b>Verification targets</b>            | [TAGS]                 | Keywords categorising the solution's purpose. Multiple selections allowed.                        |
| <b>Sector</b>                          | [TAGS]                 | Application domain(s) where the solution is relevant. Multiple selections allowed.                |
| <b>Target systems</b>                  | [TAGS]                 | Types of AI systems that the solution can verify. Multiple selections allowed.                    |

|                                                     |             |                                                                                    |
|-----------------------------------------------------|-------------|------------------------------------------------------------------------------------|
| <b>[OPTIONAL]<br/>Target legal<br/>requirements</b> | <i>Text</i> | Textual description of the identify legal requirements to address through testing. |
|-----------------------------------------------------|-------------|------------------------------------------------------------------------------------|

## 5. Mapping to the domain model

When introducing a new plugin, the plugin developer is expected to compile the following information which, together with the plugin metadata, enables mapping the AI assessment solution to the domain model; both sources are complementary and should be used in combination to fully characterise the solution.

### 4.1 Input Specification

A representative input example should be provided and explained. For instance, for an ML model this could be the main features fed to the model; for a Generative AI system, the prompting logic, prompt structure, and any system instructions.

### 4.2 Configuration Parameters

Each configuration parameter exposed by the solution should be listed and explained in terms of what it controls (e.g. thresholds, evaluation scope etc.).

### 4.3 Output Specification

A representative output example should be provided, documenting the following:

- A. Each metric — its data type, range, unit, and what it represents in the context of AI behaviour
- B. Whether each metric is direct or derived; if derived, provide the formula
- C. Any additional output fields
- D. Contextual guidance for interpreting results (e.g. how to identify hallucinations, bias signals, or unsafe outputs)
- E. How each metric addresses the AI dimensions identified in Section 3

### 4.4 Visualisation

The plot type(s) best suited to represent the AI assessment results should be identified and justified (e.g. confusion matrix, barplot, scatter plot).

### 4.5 Reporting

The information to be highlighted in a report on these results should be specified. Where an external reviewer needs to identify the specific test samples on which the AI system failed or produced unacceptable outputs, the presentation approach for that information should also be described.

## APPENDIX

This appendix reports the list of available tags for all tag-based fields in the plugin metadata template described in Section 3.

### A1. Verification targets – Testing [TAGS]

Keywords categorising the purpose of plugins for technical testing. This taxonomy draws largely from the [joint paper by LIST and RISE](#), complemented by the [OWASP Top 10 for Large Language Model Applications](#) with respect to technical robustness and safety.

| Requirement                                       | Sub-dimension                               |
|---------------------------------------------------|---------------------------------------------|
| <b>Human Agency and Oversight</b>                 | Override & opt-out mechanism testing        |
|                                                   | Audit trail & logging completeness          |
| <b>Technical Robustness and Safety</b>            | Stress testing                              |
|                                                   | Adversarial input testing                   |
|                                                   | Out-of-distribution testing                 |
|                                                   | Drift detection                             |
|                                                   | Model calibration                           |
|                                                   | Prompt Injection                            |
|                                                   | Insecure Output Handling                    |
|                                                   | Training Data Poisoning                     |
|                                                   | Sensitive Information Disclosure            |
|                                                   | Insecure Plugin Design                      |
|                                                   | Excessive Agency                            |
|                                                   | Overreliance                                |
|                                                   | Model Theft                                 |
| <b>Privacy and Data Governance</b>                | PII detection testing                       |
|                                                   | Data anonymisation / re-identification risk |
|                                                   | Data retention policy violation detection   |
| <b>Transparency</b>                               | Local interpretability                      |
|                                                   | Global interpretability                     |
|                                                   | Explanation consistency testing             |
| <b>Diversity, Non-discrimination and Fairness</b> | Bias detection on data                      |
|                                                   | Bias detection on system                    |
| <b>Societal and Environmental Wellbeing</b>       | Energy consumption tracking                 |
|                                                   | Hardware utilisation profiling              |

## A2. Verification targets – Controls [TAGS]

Keywords categorising the purpose of solutions for controls checking. This taxonomy draws largely from the [joint paper by LIST and RISE](#).

| Requirement                                       | Sub-dimension                     |
|---------------------------------------------------|-----------------------------------|
| <b>Human Agency and Oversight</b>                 | Supervisory control               |
|                                                   | Strategic governance              |
|                                                   | Informed consent                  |
|                                                   | User preferences                  |
|                                                   | Audit trails                      |
| <b>Technical Robustness and Safety</b>            | Fail-over architecture            |
|                                                   | Human fallback                    |
| <b>Privacy and Data Governance</b>                | Purpose limitation                |
|                                                   | Scope restriction                 |
|                                                   | Pseudonymisation                  |
|                                                   | Role-based access control         |
|                                                   | Lineage tracking                  |
|                                                   | Metadata versioning               |
|                                                   | Consent tracking                  |
|                                                   | Granular revocation               |
| <b>Transparency</b>                               | Model cards                       |
|                                                   | Datasheets                        |
|                                                   | AI identity notices               |
|                                                   | Disclaimers                       |
|                                                   | Risk communication                |
|                                                   | Understandable language           |
| <b>Diversity, Non-discrimination and Fairness</b> | Sampling strategies               |
|                                                   | Accessibility                     |
|                                                   | Participatory design              |
| <b>Societal and Environmental Well-being</b>      | Rights-based system               |
|                                                   | Independent ethics committee      |
| <b>Accountability</b>                             | RASCI charts                      |
|                                                   | Logging                           |
|                                                   | External reporting interfaces     |
|                                                   | Redress systems                   |
|                                                   | Version-controlled records        |
| <b>Quality Management</b>                         | Quality Policy Declaration        |
|                                                   | Control of documented information |
|                                                   | Internal Auditing                 |
|                                                   | Continuous Improvement            |
| <b>Risk Management</b>                            | Description of AI system          |
|                                                   | Structured risk identification    |
|                                                   | Core classification step          |
|                                                   | Finalise classification decision  |

|                                |                                        |
|--------------------------------|----------------------------------------|
|                                | Outline potential risk areas           |
|                                | Establishing PMM plan                  |
|                                | Perform CAPA                           |
|                                | Compliant incident reporting           |
| <b>Technical Documentation</b> | System overview and intended purpose   |
|                                | Architecture and design specifications |
|                                | Dataset provenance                     |
|                                | Structured compliance mapping          |
| <b>Record-keeping</b>          | Operational logging system             |
|                                | Version control                        |
|                                | Full lineage                           |
|                                | Retention/deletion policies            |

### A3. Target systems

Types of AI systems that the solution can verify. This taxonomy is largely based on [Hugging Face's](#) tags. Since Hugging Face categorises AI models rather than AI systems, adaptations were necessary, including the addition of categories not covered in the original, such as multi-agent systems.

| Category                                    | Sub-category / Capability            |
|---------------------------------------------|--------------------------------------|
| <b>Multimodal</b>                           | Multimodal Understanding             |
|                                             | Multimodal Generation                |
|                                             | Multimodal Retrieval                 |
| <b>Computer Vision</b>                      | Image Understanding & Classification |
|                                             | Object Detection & Segmentation      |
|                                             | Image Generation & Editing           |
|                                             | Video Understanding & Generation     |
|                                             | 3D Understanding & Generation        |
|                                             | Depth & Spatial Estimation           |
| <b>Natural Language Processing</b>          | Text Understanding & Classification  |
|                                             | Question Answering                   |
|                                             | Text Generation & Summarization      |
|                                             | Sentence & Semantic Similarity       |
|                                             | Information Extraction               |
| <b>Audio</b>                                | Speech & Audio Recognition           |
|                                             | Speech & Audio Generation            |
| <b>Knowledge &amp; Retrieval</b>            | Retrieval-Augmented Generation (RAG) |
|                                             | Semantic Search                      |
|                                             | Knowledge Graph Systems              |
|                                             | Document Intelligence                |
| <b>Recommendation &amp; Personalization</b> | Collaborative Filtering              |
|                                             | Content-Based Recommendation         |
|                                             | Hybrid & Sequential Recommendation   |
|                                             | Personalization                      |

|                                             |                                     |
|---------------------------------------------|-------------------------------------|
| <b>Predictive &amp; Analytical AI</b>       | Predictive Analytics                |
|                                             | Time Series Forecasting             |
|                                             | Risk Scoring & Assessment           |
|                                             | Causal Inference                    |
| <b>Anomaly Detection</b>                    | Anomaly & Outlier Detection         |
|                                             | Monitoring & Alerting               |
| <b>Tabular &amp; Structured Data</b>        | Tabular Classification & Regression |
|                                             | Feature Engineering & Selection     |
| <b>Decision &amp; Optimization</b>          | Decision Support Systems            |
|                                             | Constraint Optimization             |
|                                             | Simulation & Scenario Modeling      |
| <b>Reinforcement Learning &amp; Control</b> | RL & Sequential Decision-Making     |
|                                             | Robotics & Embodied AI              |
|                                             | Process Control & Automation        |
| <b>Agents &amp; Agentic Systems</b>         | Task Planning & Decomposition       |
|                                             | Tool Use & API Orchestration        |
|                                             | Memory & Context Management         |
|                                             | Multi-Agent Coordination            |
|                                             | Human-in-the-Loop Systems           |
|                                             | Autonomous Workflow Execution       |
| <b>AI Safety &amp; Governance</b>           | Content Moderation                  |
|                                             | Privacy-Preserving AI               |
| <b>Other</b>                                | Graph Machine Learning              |
|                                             | Neuromorphic & Edge AI              |
|                                             | Quantum-Classical Hybrid AI         |
|                                             | Synthetic Data Generation           |

## A4. Sector

Application domain(s) where the solution is relevant. This taxonomy is sourced from the [OECD catalogue of AI tools](#).

| Sector                |
|-----------------------|
| Agriculture           |
| Competition           |
| Corporate governance  |
| Defence               |
| Digital Economy       |
| Economy               |
| Education             |
| Employment & labour   |
| Environment           |
| Finance and insurance |
| Health                |

|                             |
|-----------------------------|
| Inclusive development       |
| Industry & entrepreneurship |
| Innovation                  |
| Investment                  |
| Other                       |
| Public governance           |
| Public sector               |
| Science & technology        |
| Social & welfare issues     |
| Tax Transport               |
| Trade                       |

## A5. License

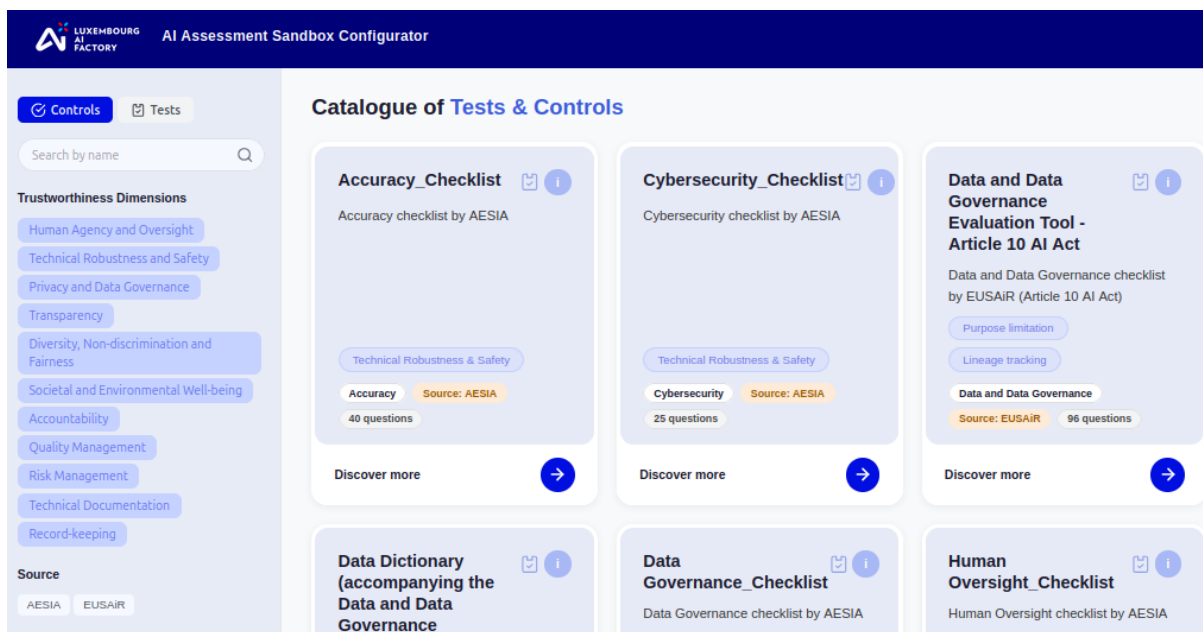
Type of license governing the use and distribution of the solution. The list of licenses is sourced from [Hugging Face](#).

| License             | Open / Closed |
|---------------------|---------------|
| proprietary         | Closed        |
| apache-2.0          | Open          |
| mit                 | Open          |
| openrail            | Open          |
| cc-by-4.0           | Open          |
| afl-3.0             | Open          |
| cc-by-sa-4.0        | Open          |
| gpl-3.0             | Open          |
| artistic-2.0        | Open          |
| bsd-3-clause        | Open          |
| agpl-3.0            | Open          |
| cc0-1.0             | Open          |
| wtfpl               | Open          |
| unlicense           | Open          |
| bsl-1.0             | Open          |
| bsd                 | Open          |
| bsd-2-clause        | Open          |
| gpl                 | Open          |
| cc-by-sa-3.0        | Open          |
| bsd-3-clause-clear  | Open          |
| cc-by-2.0           | Open          |
| cc-by-3.0           | Open          |
| cdla-permissive-2.0 | Open          |
| gpl-2.0             | Open          |
| cc-by-2.5           | Open          |
| lgpl-3.0            | Open          |

|                           |        |
|---------------------------|--------|
| osl-3.0                   | Open   |
| mpl-2.0                   | Open   |
| gfdl                      | Open   |
| pddl                      | Open   |
| ecl-2.0                   | Open   |
| ms-pl                     | Open   |
| etalab-2.0                | Open   |
| epl-2.0                   | Open   |
| cdla-sharing-1.0          | Open   |
| odc-by                    | Open   |
| cdla-permissive-1.0       | Open   |
| lgpl                      | Open   |
| eupl-1.2                  | Open   |
| eupl-1.1                  | Open   |
| odbl                      | Open   |
| lppl-1.3c                 | Open   |
| zlib                      | Open   |
| epl-1.0                   | Open   |
| lgpl-2.1                  | Open   |
| isc                       | Open   |
| ncsa                      | Open   |
| lgpl-lr                   | Open   |
| postgresql                | Open   |
| ofl-1.1                   | Open   |
| bigscience-openrail-m     | Open   |
| creativeml-openrail-m     | Closed |
| cc-by-nc-4.0              | Closed |
| llama3                    | Closed |
| gemma                     | Closed |
| llama2                    | Closed |
| openrail++                | Closed |
| cc-by-nc-sa-4.0           | Closed |
| bigscience-bloom-rail-1.0 | Closed |
| bigcode-openrail-m        | Closed |
| llama3.3                  | Closed |
| cc                        | Closed |
| cc-by-nc-nd-4.0           | Closed |
| cc-by-nc-2.0              | Closed |
| cc-by-nd-4.0              | Closed |
| cc-by-nc-3.0              | Closed |
| apple-amlr                | Closed |
| cc-by-nc-nd-3.0           | Closed |
| cc-by-nc-sa-2.0           | Closed |
| cc-by-nc-sa-3.0           | Closed |

|                                     |        |
|-------------------------------------|--------|
| apple-ascl                          | Closed |
| deepfloyd-if-license                | Closed |
| fair-noncommercial-research-license | Closed |
| c-uda                               | Closed |
| llama4                              | Closed |
| grok2-community                     | Closed |
| openmdw-1.0                         | Closed |
| intel-research                      | Closed |
| h-research                          | Closed |
| open-mdw                            | Closed |
| Other                               | NA     |

## A6. Source – Controls [TAGS]



**Figure 2. User interface of the catalogue. Users can explore controls using the tags on the right that serve as filters.**

This field identifies the originating body that authored a control checklist: the organisation whose guidance the control operationalises. It's an open, non-exhaustive taxonomy: the two sources below are the initial entries for the current release, and the list will expand as controls from further sources are onboarded. At present the controls originate from the following bodies:

| License                                                                                                                                                                     | Open / Closed                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AESIA</b> — Spanish Agency for the Supervision of Artificial Intelligence<br><a href="https://aesia.digital.gob.es/en/guides">https://aesia.digital.gob.es/en/guides</a> | 12 control checklists: Accuracy, Robustness, Cybersecurity, Data Governance, Human Oversight, Risk Management, Quality System Management, Technical Documentation, Records, Incidents Management, Post-Market Monitoring, Transparency. |

**EUSAiR project** (<https://eusair-project.eu/>)

5 control checklists: Data and Data Governance, Human Oversight, Logging, Transparency.

Each control is structured as a checklist of questions, with every question mapped to the specific EU AI Act provision(s) it addresses (e.g. Articles 9–19 and the Annex IV technical-documentation points). This per-question, article-level mapping is captured in the Controls module and complements, but is distinct from, the catalogue tag taxonomy above.